Run report: étale-cover search for T_n over every genus-2 and genus-3 curve over $\mathbb{F}_2$ (lead L8)

run-20261018T2129Z · study `research/ecdlp_leads_20261018` · commit `483bda1b`

Kind. Exact enumeration with the native Rust crate `etale_search`. It is not a timing or statistical measurement. Host: shared cloud container (Xeon 2.80 GHz, 4 cores, rustc 1.97.0), isolation L0. This run builds on run-20261018T2043Z, which covered genus 2 and hyperelliptic genus 3, and adds every smooth plane quartic. With these, every genus-3 curve over $\mathbb{F}_2$ is covered.

Method. For each curve B, $J_B(\mathbb{F}_2)$ is computed from relations: function divisors for hyperelliptic curves; for quartics, line sections, transversal conic and cubic sections, and quartic sections as a fallback. Accepting a conic, cubic or quartic section needs $4 \deg G$ distinct zeros in the factor base, so by Bézout every intersection is transversal and none is missed. The curve is accepted only with free rank 1 and torsion $= L_B(1)$. Places of degree ≤ 3 generate Pic, so this is a proof of the group (derived).

For every odd prime $n \mid \#J_B(\mathbb{F}_2)$ and every character χ of $J_B(\mathbb{F}_2)/n$, $L(B,\chi,X)$ is built and checked. The checks are the RH, the full functional equation, and for quartics an out-of-sample N_6 prediction. $L(B,\chi,X)$ is then tested for the factor $(1-c\pi X)(1-c\overline{\pi}X)$, $c \in \mu_n$, $\pi = \left(t \pm \sqrt{-7}\right)/2$, both in floating point and exactly in $\mathbb{Z}[\zeta_n]$.

quantity	genus 2	genus 3 hyperelliptic	genus 3 plane quartic
smooth models	768	6,144	10,920
validated, odd prime dividing $\#J$	520	4,128	9,152
failed validation	0	0	0
RH / functional-equation failures	0 / 0	0 / 0	0 / 0
N_6 predictions checked / failed	–	–	76,440 / 0
exact vs float disagreements	0	0	0
without an $\mathbb{F}_2$-point	0	0	0
largest odd prime $\mid L(1)$; $\mid L(-1)$	19; 19	37; 37	71; 29
characters with a T_n factor	192, all $n = 3$	576, all $n = 3$	**672, all $n = 5$**

Enumeration check (sourced). The model counts equal the known stack point counts times the order of the group of coordinate changes. Genus 2: $768 = 96 \cdot 8$, with $\#M_2(\mathbb{F}_q) = q^3$. Hyperelliptic genus 3: $6144 = 192 \cdot 32$, with $\#H_3(\mathbb{F}_q) = q^5$. Plane quartics: $10920 = 168 \cdot 65$, with $\#M_3(\mathbb{F}_q) - \#H_3(\mathbb{F}_q) = q^6 + 1$ (Bergström). The quartic models fall into 78 isomorphism classes. (Counts sourced from Bergström, J. reine angew. Math. 622 (2008) and Doc. Math. 14 (2009); not re-fetched this session.)

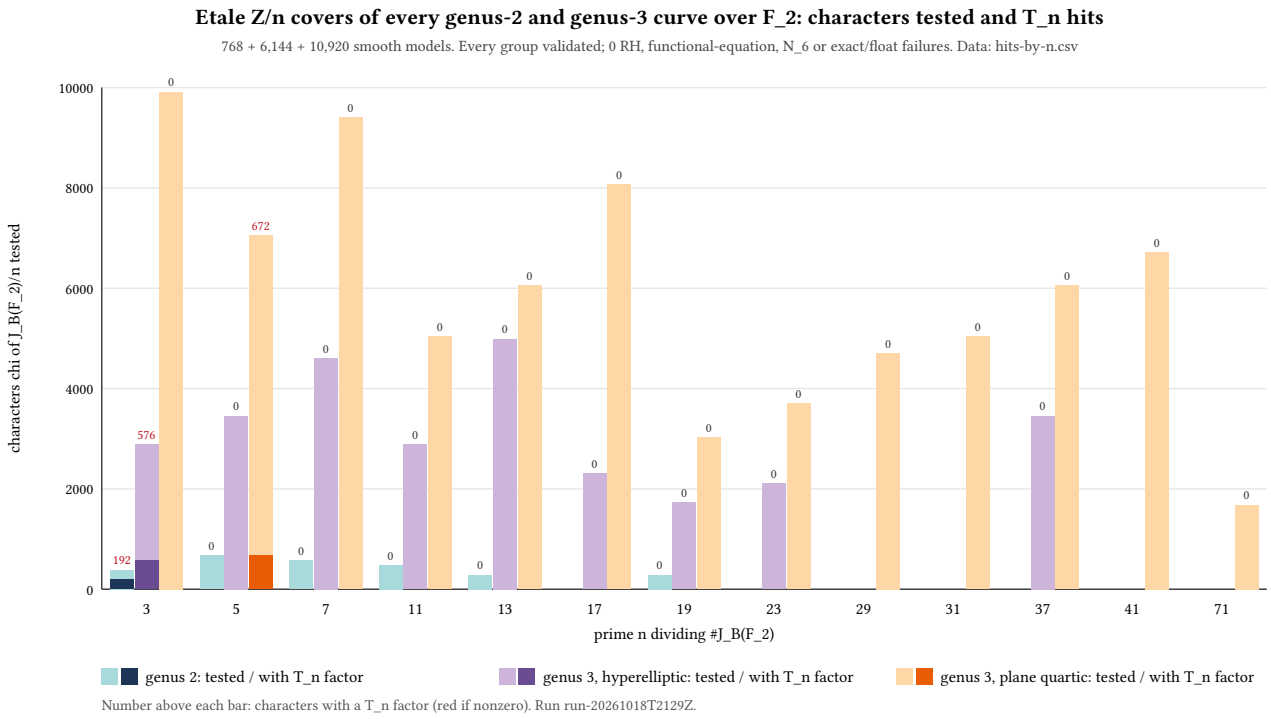


Figure 1: Characters tested and characters with a T_n factor, per n and family. Recorded values from `hits-by-n.csv`.

The $n = 5$ hit (measured). All 168 hit models form one $\mathrm{PGL}_3(\mathbb{F}_2)$-orbit. That is one curve up to isomorphism, with trivial automorphism group:

$$B : x^3y + x^2z^2 + xy^2z + xz^3 + y^3z = 0,$$

with $N = (4, 8, 19)$, $L_B(X) = 1 + X + 2X^2 + 5X^3 + 4X^4 + 4X^5 + 8X^6$ and $J_B(\mathbb{F}_2) \cong \mathbb{Z}/25$.

All four characters of $J_B/5$ give the factor with $t = -1$. That is the $a = 0$ Koblitz curve $y^2 + xy = x^3 + 1$. So, for a suitable extension of χ to $\mathrm{Pic}(B)$, the unramified $\mathbb{Z}/5$ cover $D \to B$ over $\mathbb{F}_2$ has genus 11 and T_5 inside its Prym, up to isogeny (derived from the measured factor). These are the first characters with a T_n factor for any $n \geq 5$ in this search.

Twisted covers. If Frobenius acts on the Galois group by λ of order k, then T_n in the Prym forces $k \leq g_0 - 1$ (derived). Such covers exist only if $n \mid L_B(\lambda)$. So genus 2 needs nothing beyond this search, and genus 3 adds $n \mid L_B(-1)$, recorded above.

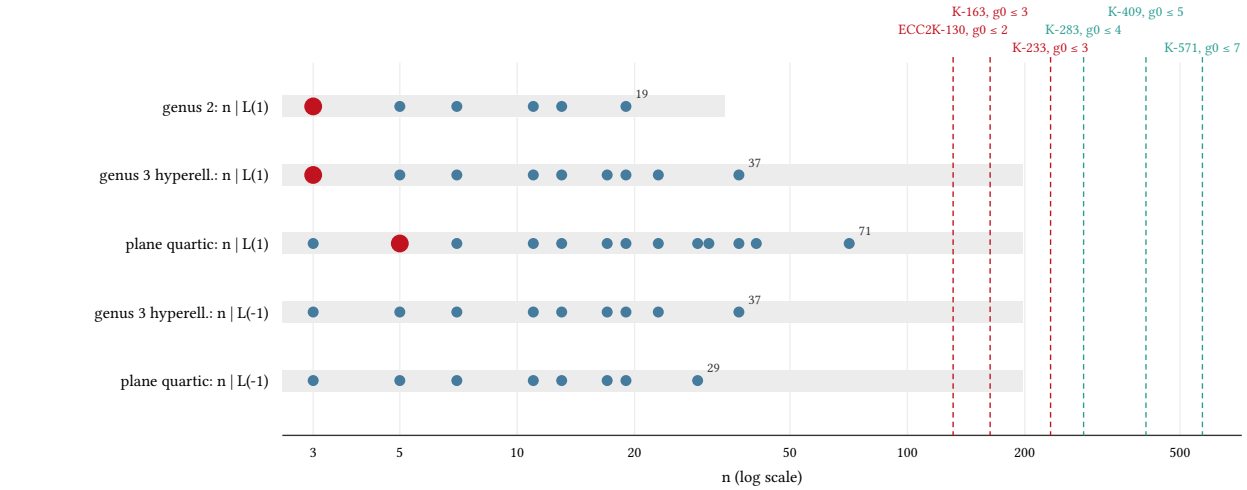


Figure 2: Odd primes n reachable by étale $\mathbb{Z}/n$ covers of genus-2 and genus-3 curves over $\mathbb{F}_2$, against the deployed Koblitz sizes and the base genus g_0 the L(1/2) budget allows (`cover-genus-budget.csv`, derived). Recorded values from `reach.csv`.

What this closes.

- ECC2K-130 needs $g_0 \leq 2$, and the Weil bound $\left(1+\sqrt{2}\right)^4 \approx 34 < 131$ rules that out (derived).
- K-163 needs $g_0 \leq 3$, and no genus ≤ 3 curve over $\mathbb{F}_2$ has $163 \mid L(\pm 1)$ (measured, exhaustive).
- K-233 needs $g_0 \leq 3$, and $\left(1+\sqrt{2}\right)^6 \approx 198 < 233$ rules that out (derived).

So the étale route is closed at these three sizes. K-283, K-409 and K-571 stay open: they need genus-4 or larger bases, with twists of order up to $g_0 - 1$.

Not covered. Genus ≥ 4 bases, ramified covers (L7), and non-cyclic covers. The novelty audit is owed.

Records. `RUN.txt` (commit, host, SHA-256 of outputs), `genus2.jsonl`, `genus3.jsonl`, `quartic.jsonl` (hits, classes, summaries), `hits-by-n.csv` and `reach.csv` (via `tables.awk`), `hits-by-n.plot.awk`, `reach.plot.awk`.